Rodolfo's Encyclopedia

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Part I

Number Theory

Chapter 1

Basic Theory

1.1 Divisibility

Definition 1. A number n is divisible by m if there exists some integer r such that $mr = n$. Also, we say m divides n and write this symbolically as

$$m \mid n$$

Theorem 2. *Let x, y, z be integers.*

- *We have $x \mid x$.*
- *We always have $1 \mid x$ and $x \mid 0$.*
- *If $x \mid y$ and $y \mid z$, then $x \mid z$.*

Theorem 3. *If $x, y \in \mathbb{Z}$, and $x \mid y$ then $y = 0$ or $\mid x \mid \leq \mid y \mid$*

Theorem 4. *Suppose $c, x, y \in \mathbb{Z}$, and $c \mid x, y$ then $c \mid ax + by$ for any $a, b \in \mathbb{Z}$*

1.2 Euclid's Division Lemma

Theorem 5 (Euclid's Division Lemma)**.** *For any integers b, a, we can find **unique** integers q, r such that*

$$b = aq + r, 0 \leq r < a.$$

*Namely, q is called the **quotient** and r the **remainder**.*

1.3 Primes

Definition 6. A natural number is called a **prime** if it has only two positive divisors, 1 and itself.

Theorem 7 (Euclid)**.** *There are infinitely many primes.*

1.4 Fundamental Theorem of Arithmetic

Theorem 8 (Fundamental Theorem of Arithmetic)**.** *Any natural number greater than* 1 *has a* ***unique*** *prime factorization upto order.Thus, any number* n *can be written as*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

Where $p_1, p_2, \dots, p_k$ *are primes and* $\alpha_1, \alpha_2, \dots, \alpha_k$ *are non-negative integers.*

Lemma 9. $\sqrt{p}$ *is irrational for any prime* p*.*

1.5 GCD and LCM

Definition 10. The GCD, or the **Greatest Common Divisor** of two numbers is the number obtained by the set of common prime factors. For two numbers m, n, it is denoted by $\gcd mn$, or often just (m, n)

Lemma 11. *Let* a*,* b *be integers. The* ***GCD*** *of* a*,* b *is the largest number which divides both* a*,* b*. In particular,* $\gcd ab \leq a, b$*.*

Lemma 12. *Let* a, b, c *be three integers. Then*

$$c \mid a, c \mid b, \Rightarrow c \mid \gcd(a, b)$$

Theorem 13 (Prime Factorization of GCD)**.** *Let* a*,* b *be two integers with prime factorization:*

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$
$$b = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$$

where α_i*,* β_i *are non-negative integers. then*

$$\gcd ab = p_1^{\min \alpha_1 \beta_1} p_2^{\min \alpha_2 \beta_2} \dots p_k^{\min \alpha_k \beta_k}$$

Definition 14. The LCM, or the **Least Common Multiple** of two numbers is the **least** natural divisible by **both** a,b. It is denoted by $\operatorname{lcm} ab$, and in particular $a,\ b \leq \operatorname{lcm} ab$.

Definition 15 (LCM)**.** Let a, b be two integers with prime factorization:

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$
$$b = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$$

where α_i, β_i are non-negative integers. then

$$\operatorname{lcm} ab = p_1^{\max \alpha_1 \beta_1} p_2^{\max \alpha_2 \beta_2} \dots p_k^{\max \alpha_k \beta_k}$$

Lemma 16. *Let* a, b, c *be three integers. Then*

$$a \mid c, b \mid c, \Rightarrow \operatorname{lcm}(a, b) \mid c$$

Theorem 17 (Product of GCD and LCM)**.** *. Let* a*,* b *be two integers. Then* $\gcd(a, b) \operatorname{lcm}(a, b) = ab$*.*

Definition 18. Two numbers a, b are called coprime or relatively prime if $\gcd(a, b) = 1$.

Lemma 19 (Euclid's Lemma)**.** *If* $c \mid ab$ *and* $\gcd ca = 1$*, then* $c \mid b$*.*

Lemma 20. *Let* a*,* m*,* n *be positive integers. Then,*

$$\gcd a^m - 1 a^n - 1 = a^{\gcd mn} - 1.$$

Lemma 21. *If* p *is an odd prime, and* a, b *are coprime, then*

$$\gcd \frac{a^p + b^p}{a + b} a + b \in \{1, p\}$$

1.6 Euclid's Division Algorithm

Lemma 22. *Let a, b be integers. We can write $a = bq + r$ for integers q, r where $0 \leq r < b$. Then the lemma states that*

$$gcd(a, b) = gcd(r, b).$$

1.7 Bézouts' Theorem

Theorem 23 (Bézouts' Theorem)**.** *Let a, b be integers. Then the equation*

$$ax + by = n$$

has a solution if and only if $\gcd ab$ *divides n.*

1.8 Chicken Mc Nuggets

Theorem 24 (Chicken Mc Nuggets Theorem/ Frobenius Coin Problem)**.** *Let a, b be coprime integers. Then the largest number that can not be expressed on the form*

$$am + bn, \ m, n \in \mathbb{Z}_{\geq 0}$$

Is $(m-1)(n-1) - 1$.

1.9 Base Systems

Definition 25. Given a positive integer b, the base$-b$ representation of a real number r is a string of digits:

$$(a_n a_{n-1} \dots a_1 a_0 . a_{-1} a_{-2} \dots)_b$$

Where $a_i \in \{0, 1, \dots, b-1\}$, and

$$r = a_n b^n + a_{n-1} b^{n-1} \dots a_1 b^1 + a_0 b^0 + a_{-1} b^{-1} a_{-2} b^{-2} \dots$$

Theorem 26. *The base$-b$ representation of any real number is unique.*

1.10 Four Number Lemma

Theorem 27 (Four number Lemma)**.** *Let a, b, c, and d be positive integers such that $ab = cd$. Show that there exists positive integers p, q, r, s such that*

$$a = pq, b = rs, c = ps, d = qr$$

Chapter 2

Basic Modular Arithmetic

2.1 Properties

2.2 Fermat's Little Theorem

Theorem 28.

Definition 29. For integers a, b and n, we write

$$a \equiv b \pmod{n}$$

if $n \mid a - b$

2.3 Residue classes

Definition 30. For a natural number n, and a non-negative number $r < n$. Then the **rth residue class** is the set of integers a that satisfy $a \equiv r \pmod{n}$. Equivalently, it is the set of all integers that leave r as a remainder when divided by n.

Theorem 31 (Properties of Modulus)**.** *Let m be a positive integer, and a, b, c, d be integers such that for a given integer n, $a \equiv b \pmod{n}, c \equiv d \pmod{n}$. Then*

- $a + c \equiv b + d \pmod{n}$
- $ac \equiv bd \pmod{n}$
- $a^m \equiv b^m \pmod{n}$

Lemma 32. *Let P be a polynomial with integer coefficients. Then*

$$P(a) \equiv P(b) \pmod{a - b}$$

2.4 Equal Sets

Theorem 33 (Two Equal Sets)**.** *Let n be an integer and consider S be the set of integers less than n and coprime to n. Let a be any integer coprime to n. Then*

$$aS \equiv S \pmod{n}$$

. Here, aS means the set obtained on multiplying each element of S by a.

2.5 Fermat's Little theorem

Theorem 34 (Fermat's Little theorem)**.** *Let a be any number coprime to a prime p. Then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for any a

$$a^p \equiv a \pmod{p}.$$

2.6 Inverses

Theorem 35 (Inverses)**.** *Let p be a prime and a be an integer coprime to p. Then there exists an integer x such that*

$$xa \equiv 1 \pmod{p}$$

*The integer x is called the **inverse** of a*

2.7 Wilson's Theorem

Theorem 36 (Wilson's Theorem)**.** *For any integer n, Then*

$$(n-1)! \equiv -1 \pmod{n} \iff n \text{ is a prime.}$$

2.8 Euler's Theorem

Definition 37. Let n be a positive integer. The function $\varphi(n)$ is called Euler's totient function, and it denotes the number of positive integers less than n that are coprime to it.

Theorem 38 (Euler's Totient Function)**.** *Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ be its prime factorization. Then*

$$\varphi(n) = n\big(1 - \frac{1}{p_1}\big)\big(1 - \frac{1}{p_2}\big) \dots \big(1 - \frac{1}{p_k}\big)$$

Equivalently,

$$\varphi(n) = n = p_1^{\alpha_1 - 1} p_2^{\alpha_2 - 1} \dots p_k^{\alpha_k - 1}(p_1 - 1)(p_2 - 1) \dots (p_k - 1)$$

Theorem 39 (φ is multiplicative)**.** *For any two **coprime** integers m, n, we have*

$$\varphi(mn) = \varphi(m)\varphi(n)$$

Theorem 40 (Euler's Theorem)**.** *Let $n \geq 2$ be an integer and a be any integer coprime to n. Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

.

Chapter 3

Arithmetic Funtions

Definition 41. An **arithmetic function**, is a function $f : \mathbb{N} \to \mathbb{C}$

Definition 42. An arithmetic function is called **multiplicative** if

$$f(mn) = f(m)f(n) \text{ for all coprime } m, n.$$

It is called completely multiplicative if

$$f(mn) = f(m)f(n) \text{ for all } m, n.$$

Definition 43. An arithmetic function is called **additive** if

$$f(mn) = f(m) + f(n) \text{ for all coprime } m, n.$$

It is called completely additive if

$$f(mn) = f(m) + f(n) \text{ for all } m, n.$$

3.1 Number of Divisors

Definition 44. Let n be a positive integer. Then the number of divisors of n as a function is denoted by $d(n)$.

Theorem 45 (Formula for $d(n)$)**.** *Let $n \in \mathbb{N}$ such that its prime factorization is*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

Then

$$d(n) = (1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_k).$$

Theorem 46. *The function $d(n)$ is odd if and only if n is a square.*

Theorem 47.

$$d(n) \leq 2\sqrt{n}$$

3.2 Sum of Divisors

Definition 48. Let n be a positive integer. Then the sum of divisors of n as a function is denoted by $\sigma(n)$.

Theorem 49 (Formula for $\sigma(n)$)**.** *Let $n \in \mathbb{N}$ such that its prime factorization is*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

Then

$$\sigma(n) = (1 + p_1 + \dots + p_1^{\alpha_1}) \dots (1 + p_k + \dots + p_k^{\alpha_k}).$$

3.3 Euler's Totient Function

Definition 50. Let n be a positive integer. The function $\varphi(n)$ is called Euler's totient function, and it denotes the number of positive integers less than n that are coprime to it.

Theorem 51 (Gauss)**.** *For any positive integer n,*

$$\sum_{d|n} \varphi(d) = n$$

Lemma 52. *For all composite n, we have*

$$\varphi(n) \leq n - \sqrt{n}$$

. Also, for all $n \neq 2, 6$,

$$\varphi(n) \geq \sqrt{n}$$

3.4 Multiplicative Functions

Theorem 53. *Let f be a multiplicative function. Then*

$$F(n) = \sum_{d|n} f(d)$$

is also multiplicative.

Lemma 54. *φ is multiplicative.*

Definition 55. The **Möbius function** $\mu(n)$ is defined as follows:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } p^2 | n \text{ for some prime } p \\ (-1)^k & \text{if } n = p_1 p_2 \dots p_k \text{ for distinct primes } p_i \end{cases}$$

Theorem 56. *$\mu(n)$ is multiplicative. Also, for any positive integer n,*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

Definition 57. The dirichlet delta function $\delta(n)$ is defined as follows:

$$\delta(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

3.4.1 Dirichlet Convolution

Definition 58. Let f and g be two arithmetic functions (not necessarily multiplicative). Then the Dirichlet convolution of f and g is defined as

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right)$$

Theorem 59. *The following properties hold for Dirichlet convolution:*

1. $f * g = g * f$
2. $f * (g + h) = f * g + f * h$
3. $(f * g) * h = f * (g * h)$
4. $f * \delta = f$

Lemma 60.

$$\sigma(n) = \sum_{m|n} \varphi(m)\, d\left(\frac{n}{m}\right)$$

3.4.2 Möbius Inversion

Theorem 61 (Möbius Inversion)**.** *Let f and g be two arithmetic functions such that*

$$g(n) = \sum_{d|n} f(d)$$

Then

$$f(n) = \sum_{d|n} g(d) \mu\left(\frac{n}{d}\right)$$

Equivalently, if $g = f * 1 \iff f = g * \mu$.

3.5 Floor and Ceiling Functions

Chapter 4

Binomial Coefficients

Definition 62. For non-negative integers a, b, we define

$$\binom{a}{b} = \begin{cases} 0 \text{ if } a < b \\ \dfrac{a!}{b!(a-b)!} \end{cases}$$

Lemma 63. *If p is a prime, and $0 < k < p$, then $p \mid \binom{p}{k}$*

4.1 Lucas's Theorem

Theorem 64 (Lucas's Theorem)**.** *Let p be a prime and m, n non-negative with base-p representations $m_k p^k + m_{k-1} p^{k-1} + \cdots + m_1 p + m_0 = m$, and $n_k p^k + n_{k-1} p^{k-1} + \cdots + n_1 p + n_0 = n$ respectively. Then,*

$$\binom{m}{n} \equiv \binom{m_k}{n_k}\binom{m_{k-1}}{n_{k-1}} \cdots \binom{m_1}{n_1}\binom{m_0}{n_0} \pmod{p}$$

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